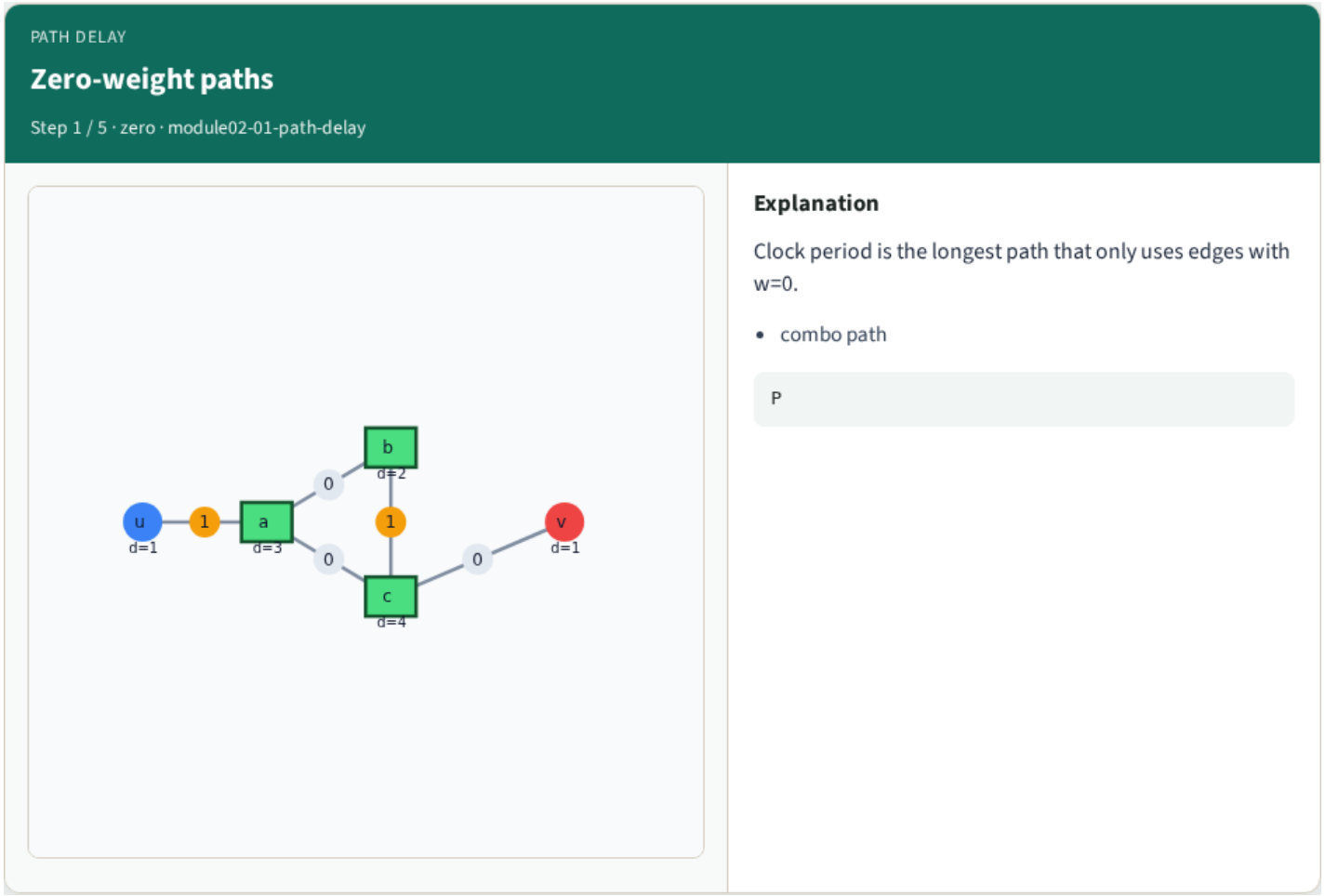
Path delay

Clock period is the longest delay along paths that use only w=0 edges (registers break paths)

The idea

- Zero edges e1, e3, e4 form comb segments; longest chain a→b→c hits delays 3+2+4 → period 8

Zero-weight paths



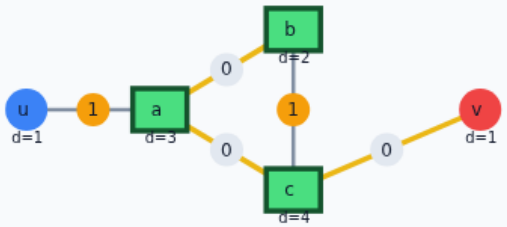
Zero-weight paths

Hot path

PATH DELAY

Hot path

Step 2 / 5 · hot · module02-01-path-delay



Explanation

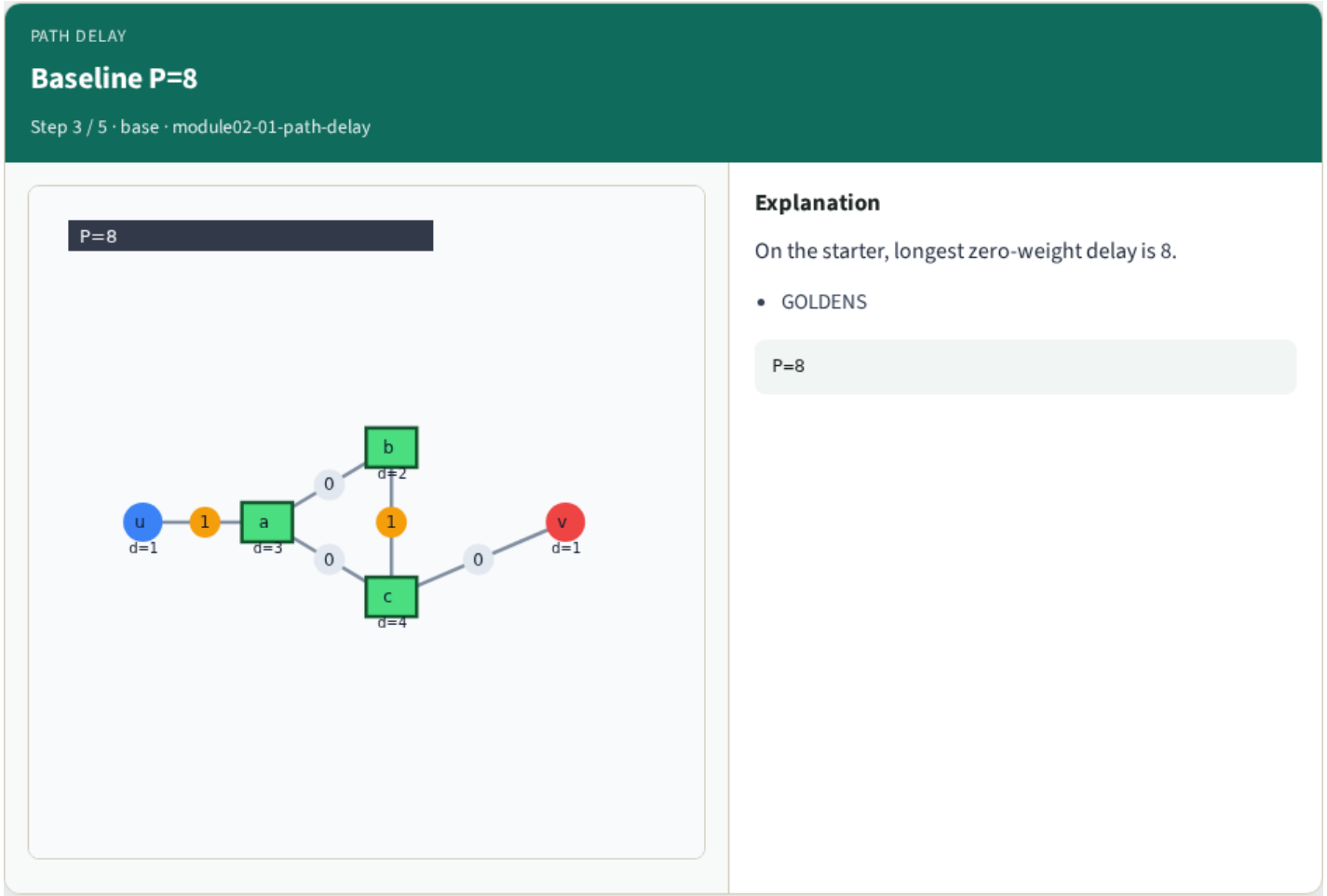
a → b and a → c and c → v create long zero-weight stretches.

- w=0

```
a b c
```

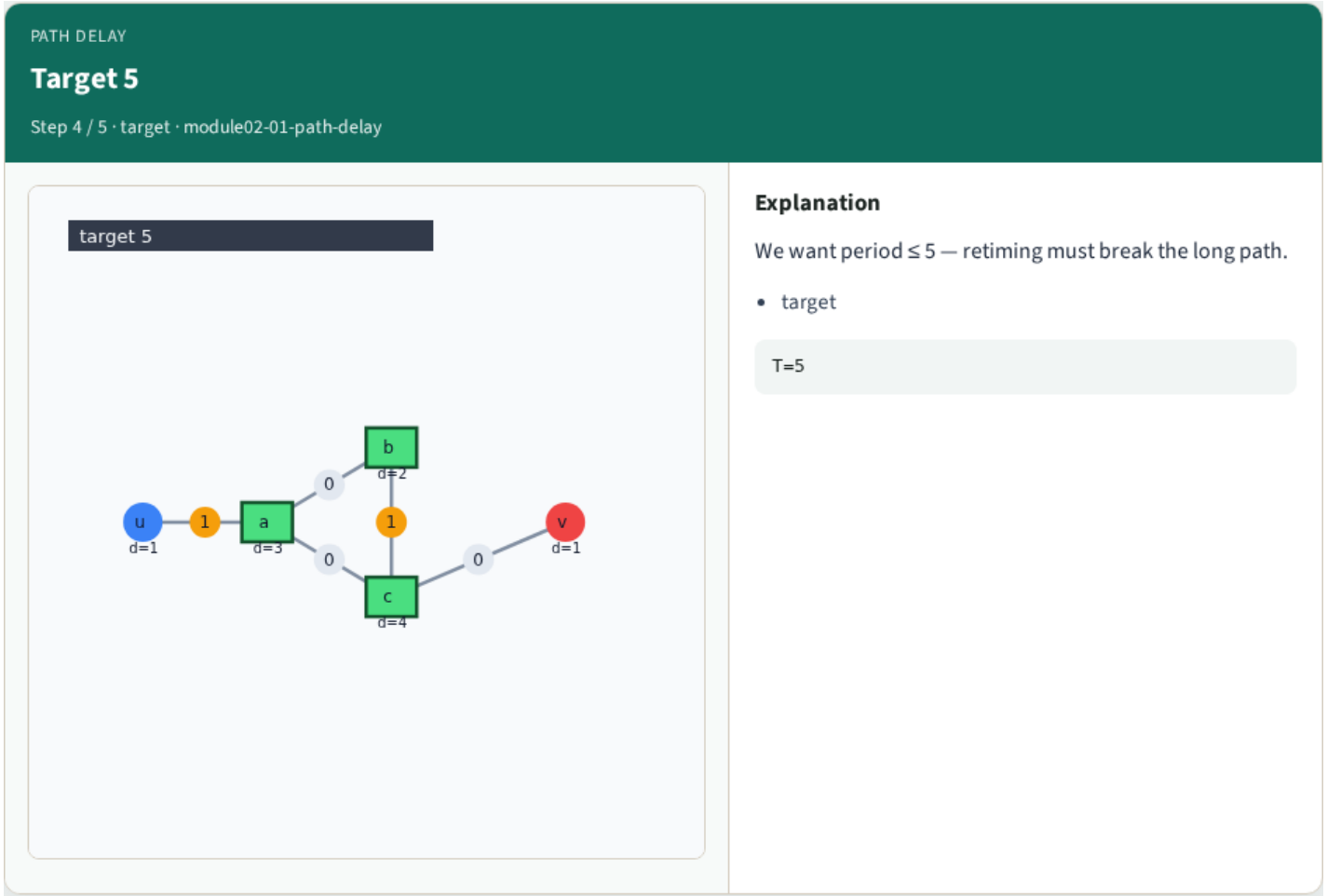
Hot path

Baseline P=8



Baseline P=8

Target 5



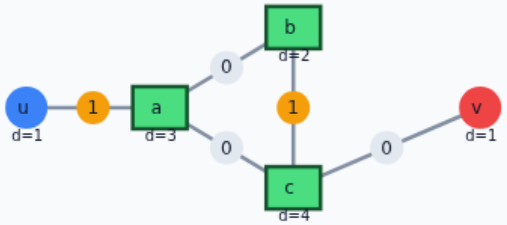
Target 5

Break with registers

PATH DELAY

Break with registers

Step 5 / 5 · idea · module02-01-path-delay



Explanation

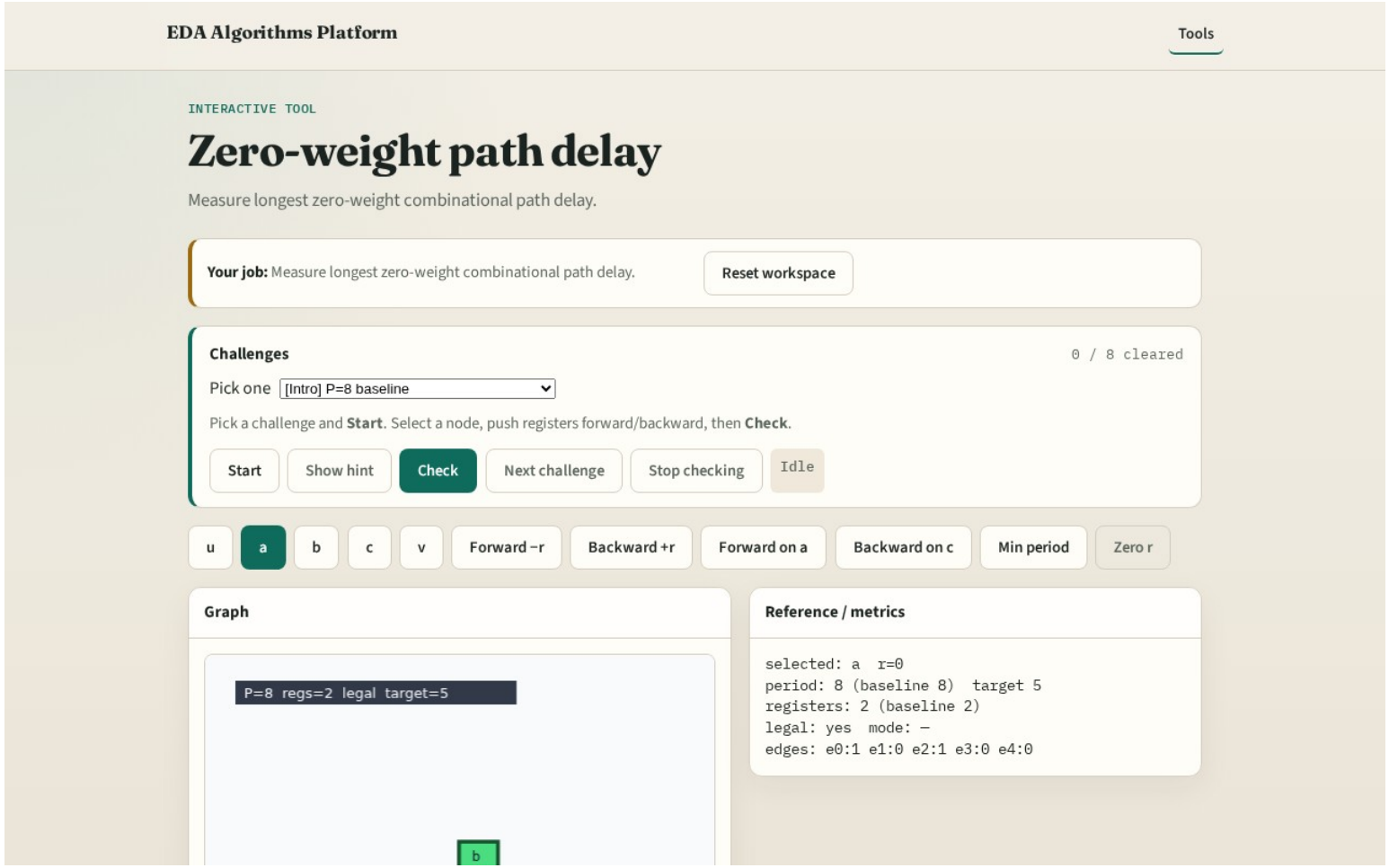
Pushing a register onto a zero-weight edge splits the combo path.

- forward/back

```
Next labs
```

Break with registers

Browser lab track



Browser lab starter

Implement track

- Call clock_period(data, zero_retime(data)); assert ≈ 8

Pitfalls

- Including w=1 edges in the critical path
- Summing edge delays instead of node delays

Your turn

- Next: forward retiming on node a